the cells that contain receptor molecules specific for the hormone are physiologically activated. A single hormone may have multiple effects on a single target tissue or on different target tissues.

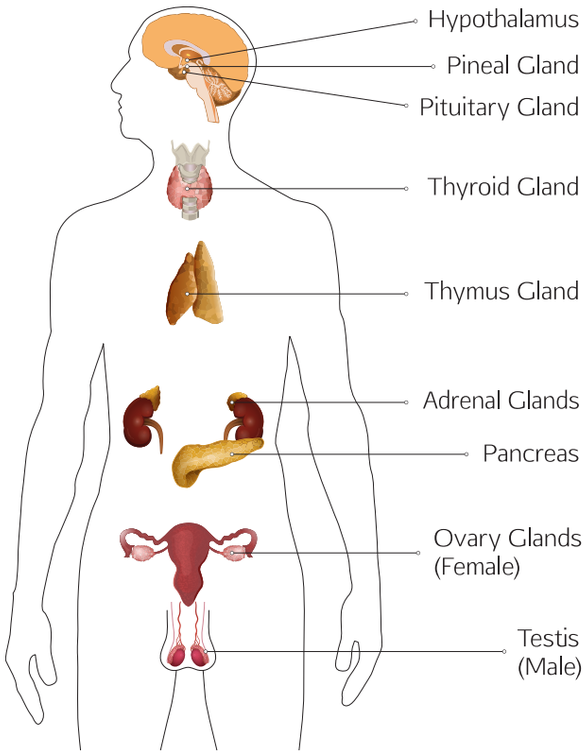


Figure 11.1 Location of various endocrine glands

Many hormones exhibit long term changes like growth, puberty and pregnancy. Hormones often influence many organs and organ systems at the same time. Serious deficiency or excess secretion of hormones leads to disorders. Hormones coordinate different physical, physiological, mental activities and maintain **homeostasis.** Hormones are composed of water soluble proteins or peptides or amines or fat soluble steroids.

Homeostasis: Maintenance of constant internal environment of the body by the different coordinating system.

11.2 Human endocrine system

There are two glandular systems such as the exocrine glands and the endocrine glands. The exocrine glands secrete enzymes, saliva and sweat and have ducts that carry their substances to the membrane surfaces. Example: salivary gland and gastric gland. The endocrine glands, called ductless glands produce hormones and lack ducts; they release their hormone to the surrounding tissue fluid. The hormones circulate around the body and eventually reach the target organs. Endocrine glands (Figure: 11. 1) include the pituitary, thyroid, parathyroid, pineal, adrenal, thymus and are also known as **exclusive endocrine glands**. The hypothalamus along with its neural function also produces hormones and is considered as a **neuro endocrine gland.** In addition several organs such as pancreas, gastro intestinal tract epithelium, kidney, heart, gonads and placenta are also have endocrine tissues and are known as **partial endocrine glands.**

Table 11.1. Chemical nature of hormones

Class	Chemical properties	Example
Amines	Small, water soluble derived from tyrosine or tryptophan	Adrenalin, nor adrenalin, melatonin and thyroid hormone
Protein/Peptides	Water soluble	Insulin, glucagon and pituitary hormones
Steroids	Derived from cholesterol mostly lipid soluble	Cortisol, aldosterone, testosterone, oestrogen, progesterone.

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